

Stone Chou

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EDUCATIONAL BACKGROUND

The State University of New York at Stony Brook (SBU)

Bachelor of Science in Physics & Mathematics

GPA: 3.85/4.0 — Graduate Physics Course GPA: (4.0/4.0)

Honors: Fumikato Fellowship (summer 2025)

Stony Brook, NY

08/2024 – 5/2027

Xi'an Jiaotong University

Honor college in the Department of Physics

GPA: 3.91/4.0

Honors: Honor college (Qian Xueseng College)

Xi'an, China

09/2023 – 06/2024

Young Gifted Program of Xi'an Jiaotong University

Jiangsu, China

selected from ninth grade to study in the University directly from Suzhou High School[SHS] 09/2022

At the age of 15.

Representative Courses

1. graduate Electrodynamics(4.0/4.0)
2. Quantum Field Theory 1,2(4.0/4.0)
two semester course by A.Zamolodchikov and P.Meade
3. Graduate Group Representation Theory(in progress)
by Peter.V.Nieuwenhuizen
4. Quantum Hall Effect and Chern Simons Theory(audit)
by Yinchen He

SELECTED PROJECTS

Atmospheric Neutrino Reconstruction in Time Projection Chamber Stony Brook, NY

—*Instructor: Changkee Jung and Ciro Riccio*

01/2025 – now

- Developing reconstruction algorithms for atmospheric neutrino events in Liquid Argon Time Projection Chambers (LArTPCs). LArTPCs can detect energy deposition above a 500 keV threshold, providing a new scenario for reconstructing neutrino energy and direction. A first-author paper based on this research is expected by the end of the year.

Topics in Quantum Many-Body Theory and Random Matrix Theory Stony Brook, NY

Undergraduate Research Project — Instructor: Jacobus Verbaarschot, SBU

08/2025 –

- Independent study in Random Matrix Theory, Quantum Many-Body Theory and SYK Model.
- Currently, we are interested in overlapping clusters SYK. I am working on calculating the probability distribution for the size of a time-evolving operator numerically and analytically in this model and determining the Lyapunov coefficients for finite N.

DUNE FD3 Photon Detector Upgrade

Instructor: Ciro Riccio

Stony Brook, NY

10/2024—10/2025

- Conducting chemical deposition of wavelength-shifting material for a next-generation VUV light trap photodetector. Testing new materials in a double-layer acrylic shifter and constructing a prototype photon detector for DUNE.

Cryogenic Test of Front-End Motherboards (FEMBs) for DUNE

Brookhaven

National Laboratory

Instructor: Shanshan Gao, Vladimir Tishchenko, BNL

- This Cold Electronics (CE) system will be implemented on Far Detectors 1 and 2 and the Vertical Drift Experiment (VD) in DUNE. It is mounted near the detector electrodes to amplify, shape, digitize, and multiplex charge signals. The QC process includes testing at room temperature (warm test), testing at cryogenic temperature (cold test), and analysis of the resulting data.

04/2025 – 08/2025

SKILL, HOBBIES & ACTIVITIES

Language Skills: Chinese (native), English (fluent)

1570/1600 in SAT test

110/120 in TOEFL test

Programming Skills: Mathematica, C++, Fortran, Python, LaTeX, Linux.

Hobbies:

Soccer (Physics department Chairman nominated **best player** in the physics department)

Argentina Tango(**Vice President** of Tango Club in Stony Brook University)

table tennis

Literature(Columnist of Boston Book of Review in 2024)

Affiliations

Neutrino and Nucleon Decay Group at Stony Brook University